CECS 361 Spring 2026

Stage 1: Proposal

Custom 16-bit RISC SoC for Digital Lock with VGA Status Display

Justin Natividad, Gabriel Dao, Mikko

Taitague, Matthew Orendain

Wednesday, March 18, 2026

Abstract

This project proposes the design and implementation of a custom 16-bit RISC system on chip on the Nexys A7 100-T Board. The system will include a simple processor core, instruction ROM, data RAM and memory-mapped peripherals for switches, push buttons, LEDs, seven segment displays, and VGA output. The processor will execute a digital lock application in which a user enters a code through board inputs, and the system responds by displaying lock status messages such as locked, access granted, access denied, and lockout. The purpose of this project is to demonstrate processor design, memory interfacing, peripheral control, and hardware/software interaction in a visible and interactive embedded environment. The completed design will be verified through simulation of individual modules and demonstrated on the Nexys A7 board with real time visual output.